

Explainable AI-Driven Fuel Efficiency Prediction Using Ensemble Learning and Vehicle Configuration Data

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Abstract

Predicting fuel efficiency is an emerging research domain with the rapid rise of fuel usage, environmental degradation and the rising demand for sustainable transport infrastructure. The precise computation of the fuel efficiency can assist the users in saving the fuel costs and analyzing the performance of vehicles in an environment friendly way. Traditional statistical approaches often struggle to model complex nonlinear relationships among vehicle parameters. To tackle this problem, this study proposes a Predictive Fuel Efficiency framework using Explainable AI and Machine Learning algorithms. In this proposed system, Auto MPG dataset and other data related to the specifications of cars are used. Data preprocessing involved handling missing values, type conversion, and removal of inconsistent records. Also feature engineering was done by computing power to weight ratio, age of the car, engine efficiency indicators and displacement. The system utilizes models like Linear Regression, Random Forest Regressor and XGBoost Regressor, and they are evaluated using metrics like R^2 Score and RMSE.

The experimental study shows that the use of XGBoost model is more effective than the use of other models due to the ability to model nonlinear functions and reduce the prediction errors. In addition to prediction of the efficiency, we propose the use of Explainable AI techniques to increase interpretability by analyzing the importance of features. Furthermore, the system estimates the environmental impacts in terms of CO₂ emissions based on the predicted fuel efficiency values of the vehicles. Overall, the framework offers a user-friendly way of fuel efficiency prediction and analysis. The research proposal can be extended by using the data of the vehicle sensors, deployment to the cloud and use of the deep learning algorithms.

Keywords— Fuel Efficiency Prediction, Explainable AI, XGBoost, Ensemble Learning, Machine Learning, Environmental Sustainability, CO₂ Emission Analysis, Vehicle Analytics.

1. Introduction

The fast development of the automobile industry and the increasing dependence on transportation systems have been major contributors to fuel consumption and environmental pollution worldwide [1], [2]. Due to rising fuel prices, increasing carbon emissions, and growing concerns regarding climate change, researchers and industries are focusing on the development of intelligent and sustainable

transportation solutions [3]. In this context, fuel efficiency prediction has become an important research area as it helps to analyze vehicle performance, reduce fuel consumption, and support environmentally responsible decision-making [1], [3].

Fuel efficiency is usually measured in miles per gallon (MPG) or kilometers per liter (KM/L). It depends on various vehicle-related factors such as engine size, horsepower, vehicle weight, acceleration, number of cylinders, and vehicle age [7], [14]. Traditional statistical methods often struggle to capture the complex nonlinear relationships among these factors, resulting in lower prediction accuracy and reduced interpretability [5], [14]. As a result, machine learning techniques have gained significant popularity because of their ability to identify hidden patterns in data and generate more accurate predictions [1],[15].

Recent advancements in Artificial Intelligence (AI) and ensemble learning techniques have enabled the development of efficient predictive systems for automotive applications [3], [8]. Algorithms such as Random Forest, Gradient Boosting, and XGBoost have demonstrated strong performance in regression-based prediction tasks due to their capability to handle nonlinear relationships and minimize prediction errors [4], [5], [6]. However, many existing systems primarily focus on prediction accuracy while offering limited transparency regarding the influence of vehicle parameters on prediction outcomes. Additionally, several current approaches do not incorporate environmental impact analysis or user-oriented intelligent prediction frameworks [2], [3].

To address these limitations, this research introduces an Explainable AI-driven fuel efficiency prediction framework using ensemble machine learning techniques and vehicle configuration data. The proposed system utilizes the Auto MPG dataset [13] along with additional vehicle specification data for model training and evaluation. Data preprocessing and feature engineering techniques are applied to improve dataset quality and model performance. Multiple machine learning algorithms including Linear Regression, Random Forest Regressor, and XGBoost Regressor are implemented and compared using performance metrics such as R^2 Score and Root Mean Squared Error (RMSE) [6], [7].

The proposed framework integrates Explainable AI techniques to improve model interpretability through feature importance analysis [8]. It also performs environmental impact assessment by estimating CO₂ emissions based on predicted fuel efficiency values. Additionally, an interactive prediction framework has been developed to allow users to analyze fuel efficiency predictions using vehicle configuration parameters. The proposed approach aims to provide an intelligent, transparent, and sustainability-oriented solution for transportation analysis and decision-making.

The major contributions of this work are:

- Designing an Explainable AI fuel efficiency prediction system.
- Evaluating various machine learning and ensemble learning techniques.
- Incorporating CO₂ emission estimation in the prediction process.
- Designing an interactive dashboard for prediction and simulation.

2. Problem Statement

The prediction of fuel efficiency plays a crucial role in the automotive sector since it determines the amount of fuel consumed and the environmental impact of the vehicles. The existing methods for predicting the fuel efficiency of the cars include statistical analysis and mathematical modeling, which may not be able to capture the complex interactions between various vehicle factors and fuel efficiency. Machine learning algorithms are highly efficient and reliable in terms of predicting vehicle fuel efficiency; however, most of them lack transparency. In other words, people may not be able to comprehend how certain factors affect their predictions of vehicle fuel efficiency. Thus, there is a need for an intelligent prediction system that will be capable of predicting the fuel efficiency of vehicles and analyzing the impact of the use of certain fuels on the environment. The problem statement for the current project is as follows: Development of an Explainable Artificial Intelligence-based model to predict vehicle fuel efficiency using an ensemble of machine learning models.

3. Objectives

The main goal of this research is to create an efficient fuel efficiency prediction system using machine learning and ensemble learning techniques. The study seeks to explore the link between vehicle parameters and fuel consumption to improve prediction accuracy and support sustainable transportation assessments. The proposed system aims to include Explainable AI techniques to boost model transparency and help users understand how different vehicle features affect fuel efficiency predictions.

The specific objectives of this research are:

- To predict vehicle fuel efficiency using machine learning algorithms like Linear Regression, Random Forest, and XGBoost.
- To compare the performance of ensemble learning models using evaluation metrics such as R^2 Score and Root Mean Squared Error (RMSE).
- To enhance prediction accuracy through feature engineering methods including power-to-weight ratio, vehicle age, and engine efficiency indicators.
- To incorporate Explainable AI techniques to improve model interpretability and feature importance analysis.
- To assess the environmental impact of vehicles by analyzing CO₂ emissions based on predicted fuel efficiency values.

4. Literature Survey

Vehicle fuel efficiency prediction has emerged as an important research area due to increasing fuel consumption, environmental concerns, and the growing demand for sustainable transportation systems. Traditional statistical models often struggle to capture nonlinear relationships among the parameters involved. Because of this, researchers are turning to machine learning for predictions, as these models can better handle nonlinearity [1], [2], [3]. Recent studies have demonstrated the effectiveness of machine learning algorithms in improving fuel efficiency prediction accuracy. For example, Yoo et al. [1] developed a machine learning framework that used various vehicle features and tested several regression models to predict fuel efficiency. They found that employing ensemble learning led to better predictions. Another study by Zhang et al. [2] applied machine learning models to actual driving data while considering vehicle and operational characteristics in fuel efficiency predictions.

Ensemble methods have become very popular among prediction algorithms. They are robust and reduce prediction errors. The Random Forest algorithm [4], proposed by Breiman, creates multiple decision trees to enhance predictive power and avoid overfitting. Friedman [5] introduced gradient boosting, which iteratively improves predictions and reduces prediction errors. The XGBoost algorithm [6], an enhanced version of gradient boosting, is known for its scalability, efficiency, and quality of predictions for regression tasks.

The literature provides a detailed understanding of the theoretical foundations of predictive modeling and supervised learning. Works by James et al. [7] and Hastie et al. [14] discuss machine learning techniques and their evaluation. Raschka and Mirjalili [15] offer practical applications of these machine learning approaches.

Explainable Artificial Intelligence (XAI) is a growing research area focused on making machine learning models more transparent and interpretable. Molnar [8] presents various ways to use model explanations to assess the impact of input variables on predictions. Similarly, Dwivedi et al. [18] highlight XAI's role in fostering trust, accountability, and transparency in AI systems. These explainability approaches improve our understanding of prediction outcomes and help in making informed decisions.

Meanwhile, several researchers have explored how driving behavior and vehicle conditions affect fuel consumption. For instance, Rios-Torres et al. [19] demonstrated that driving style significantly

influences efficiency and environmental impact. Additionally, Dubois [17] emphasized predictive analytics through AI solutions for smart vehicles. Studies by Alm'er [20] have shown that machine learning techniques are effective in predicting fuel consumption in heavy vehicles.

Various software packages and datasets have facilitated the creation of fuel efficiency prediction systems using machine learning. The Scikit-learn library [9], Pandas package [11], and general-purpose data science tools in Python [10] are common choices for data preparation and model development. Streamlit [12] helps build interactive machine learning applications, and the Auto MPG dataset [13] remains a frequently used benchmark in vehicle fuel efficiency studies.

4.1. Research Gap Analysis

Despite previous studies successfully enhancing the accuracy of fuel efficiency predictions with machine learning, many focus primarily on predictive accuracy. There is limited exploration of Explainable AI, environmental impact analysis, and user-centered prediction systems. Most approaches lack adequate sustainability analysis. Therefore, this study integrates ensemble machine learning algorithms, Explainable AI methods, CO₂ emission estimation, and an interactive dashboard approach into a single system to improve fuel efficiency predictions.

5. Proposed Methodology

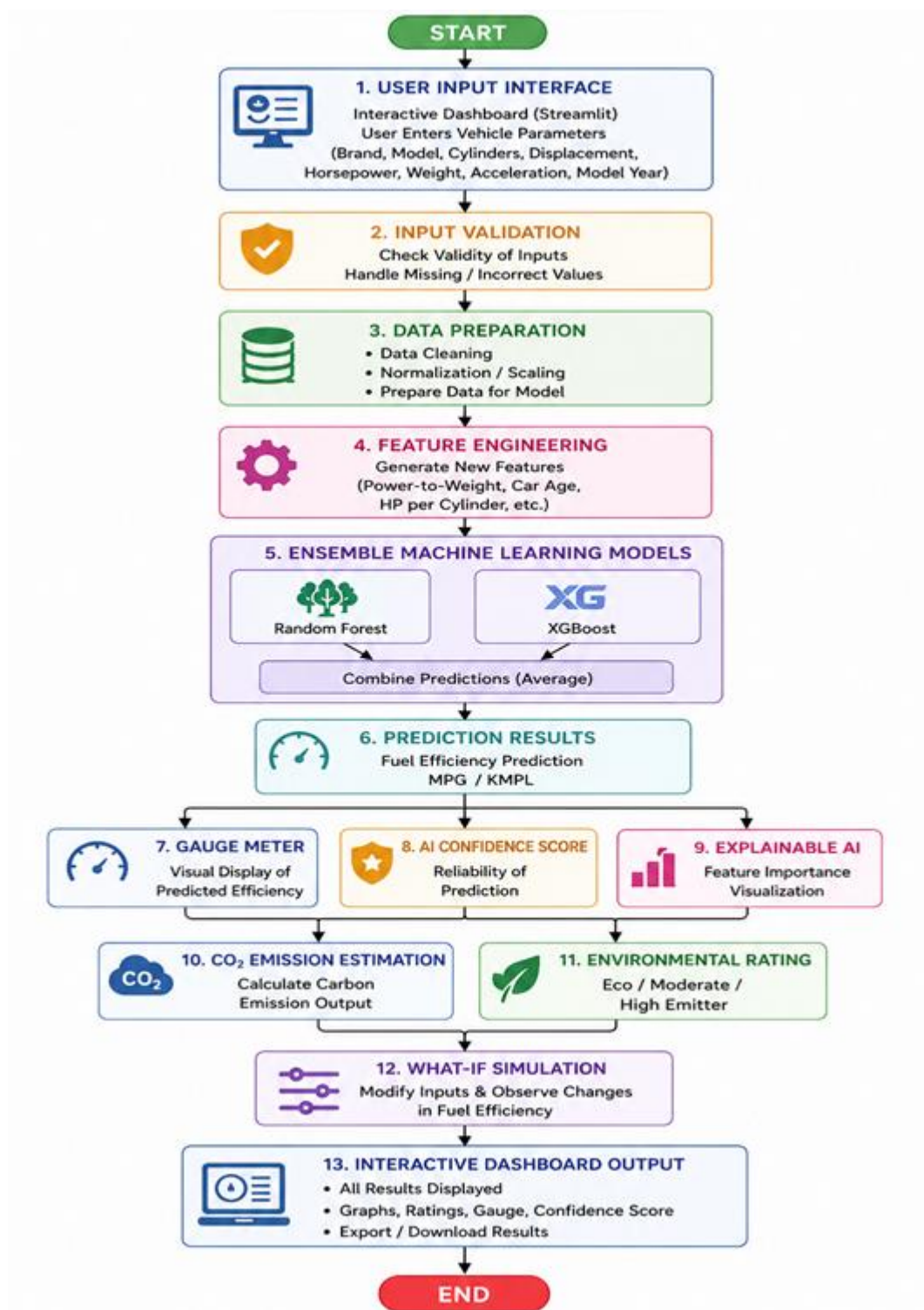


Fig. 1. Proposed System Workflow

The proposed workflow consists of data acquisition, preprocessing, feature engineering, model training, Explainable AI analysis, CO₂ emission estimation, and dashboard-based prediction. The system predicts vehicle fuel efficiency using Explainable AI and ensemble machine learning techniques [1], [2], [3]. The workflow of the framework is shown in Fig. 1. First, users provide vehicle parameters such as brand, model, horsepower, cylinders, weight, acceleration, and model year through an interactive

dashboard [12]. The input data undergoes validation and preprocessing using data cleaning, normalization, and scaling methods [7], [9], [11].

After preprocessing, feature engineering techniques create meaningful features like power-to-weight ratio, vehicle age, and horsepower per cylinder. Ensemble learning models, including Random Forest and XGBoost, are used for prediction and evaluated with R² Score and RMSE metrics [14]. The system integrates Explainable AI for feature importance analysis [8] and estimates CO₂ emissions to assess environmental impact [3]. Additionally, a What-If simulation module allows users to change vehicle parameters and see how predictions change through the interactive dashboard [12].

6. Dataset Description

The proposed fuel efficiency prediction system uses two datasets for model training and analysis. The main dataset is the Auto MPG dataset from the UCI Machine Learning Repository [13]. This dataset includes vehicle-related parameters for predicting fuel efficiency.

6.1. Auto MPG Dataset

The key features taken from the dataset are Cylinders, Horsepower, Weight, Acceleration, Model Year

6.2. Car Specifications Dataset

We also use a car specifications dataset to assist with vehicle-based prediction and analysis. The key features from this dataset are Brand, Model, Engine HP, Cylinders, Fuel Type

The combination of both datasets improves prediction performance and supports real-world vehicle analysis.

Table 1 DATASET DESCRIPTION TABLE

Dataset	Records	Features
Auto MPG	398	MPG, Cylinders, Horsepower, Weight, Acceleration, Model Year
Car Specifications	11816	Brand, Model, Engine HP, Fuel Type, Cylinders

7. Data Preprocessing & Feature Engineering

The collected datasets had missing values, inconsistent records, and different data formats, which had to be pre-processed before model training. Data cleaning techniques were used to clean the data by removing invalid records and handling missing values. Model performance was enhanced and data variability reduced by standardizing numerical data types and normalization techniques [11].

Feature engineering was performed to generate useful features to improve prediction accuracy. Features like power-to-weight ratio, vehicle age, horsepower per cylinder and average MPG were computed using existing vehicle parameters. These engineered features were very helpful for the machine learning models to identify the hidden relationships between vehicle features and fuel efficiency [1], [15].

The mathematical representation of the engineered features is given as follows:

1. Power-to-Weight Ratio

$$\text{power to weight ratio} = \frac{\text{Horsepower}}{\text{Vehicle Weight}}$$

2. Vehicle Age

$$\text{Vehicle Age} = \text{Current Year} - \text{Model Year}$$

3. Horsepower per Cylinder

$$\text{HP per Cylinder} = \frac{\text{Horsepower}}{\text{Number of Cylinders}}$$

4. Average MPG

$$\text{Average MPG} = (0.55 \times \text{Highway MPG}) + (0.45 \times \text{City MPG})$$

These engineered features improved the learning ability of the machine learning models and led to better performance in fuel efficiency prediction.

8. Model Training and Evaluation

The dataset was then split into training and testing sets following an 80:20 split ratio after the data preprocessing stage and feature creation. Various machine learning algorithms including Linear Regression, Random Forest Regressor, Gradient Boosting Regressor, and XGBoost Regressor were built to predict the efficiency of the vehicles [4], [5], [6]. These models were trained using various car attributes such as horsepower, weight, number of cylinders, acceleration, and created attributes.

Linear Regression was the regression model used as the benchmark. Algorithms including Random Forest and Gradient Boosting were implemented to boost the prediction ability using ensemble learning technique. In addition, the use of XGBoost algorithm was done because of its ability to capture nonlinearity, prevent overfitting, and increase accuracy.

Gradient boosting builds an additive model sequentially:

$$F_M(x) = \sum_{m=1}^M \eta h_m(x)$$

where w_j are regression trees, η is the learning rate, and each new tree corrects residual errors from the current ensemble.

XGBoost optimizes a regularized objective over additive trees:

$$L = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{m=1}^M \Omega(h_m)$$

with a common regularization form

$$\Omega(h) = \gamma(\text{leaves}) + \frac{\lambda}{2} \sum_j w_j^2$$

where w_j are leaf scores, γ penalizes excessive splitting, and λ shrinks leaf weights

In this case, supervised learning approach was applied in building the machine learning models from the given dataset. The input attributes of the dataset were mapped to the output attribute MPG (miles per gallon) [7], [14].

Model performance was analyzed using evaluation parameters like R² score and root mean square error (RMSE). The formula used for calculating the above-mentioned parameters is shown below:

1. R² Score

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

where:

y_i = actual value

$\hat{y}_i$ = predicted value

$\bar{y}$ = mean of actual values

2. Root Mean Squared Error (RMSE)

$$RMSE = \sqrt{\frac{1}{n} \sum (y_i - \hat{y}_i)^2}$$

where:

n = total number of observations

The comparison of trained models allowed the selection of an appropriate algorithm for fuel efficiency prediction. According to experimental results, machine learning methods including XGBoost were more efficient than other regression algorithms in terms of low error rate.

9. Results and Discussion

The performance of the developed machine learning models was evaluated using R² Score and Root Mean Squared Error (RMSE) to predict fuel efficiency. Four algorithms were trained and compared with the prepared dataset. These algorithms include Linear Regression, Random Forest, Gradient Boosting, and XGBoost.

The experimental results show that ensemble learning algorithms performed better than traditional regression methods. Among all the models, XGBoost achieved the highest R² Score and the lowest RMSE value. This demonstrates its ability to model complex nonlinear relationships and minimize prediction errors.

Table 2 MODEL COMPARISON TABLE

Model	R ² Score	RMSE
Linear Regression	0.6635	3.3864
Random Forest	0.8729	2.0810
Gradient Boosting	0.7872	2.6929
XGBoost	0.8761	2.0551

Linear Regression showed the lowest prediction accuracy because it assumes a linear relationship between the input variables and fuel efficiency. On the other hand, Random Forest improved performance by combining multiple decision trees. This allowed the model to capture complex interactions among vehicle attributes. XGBoost further improved prediction accuracy through gradient boosting and regularization methods, which helped reduce overfitting and enhanced model generalization. As a result, XGBoost provided the best overall performance with an R² Score of 0.8761 and an RMSE of 2.0551.

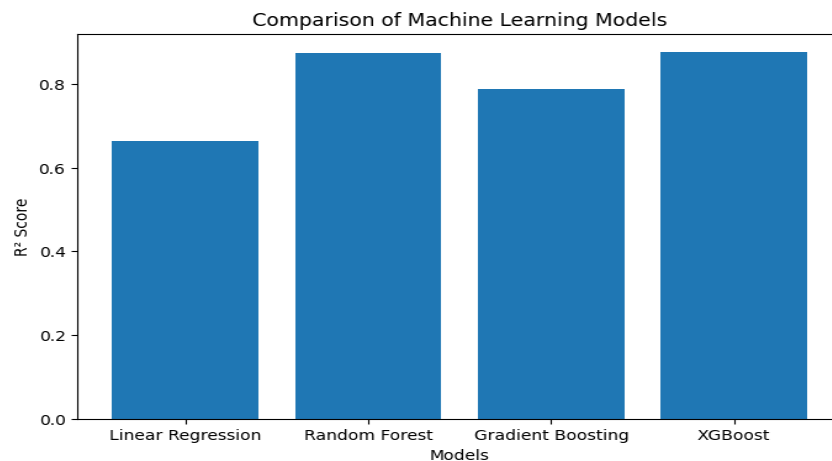


Fig. 2. Comparison of Machine Learning Models

The comparison in Fig. 2 shows that ensemble learning algorithms outperformed traditional regression techniques. XGBoost had the highest R² Score, indicating it could better explain variations in fuel efficiency.

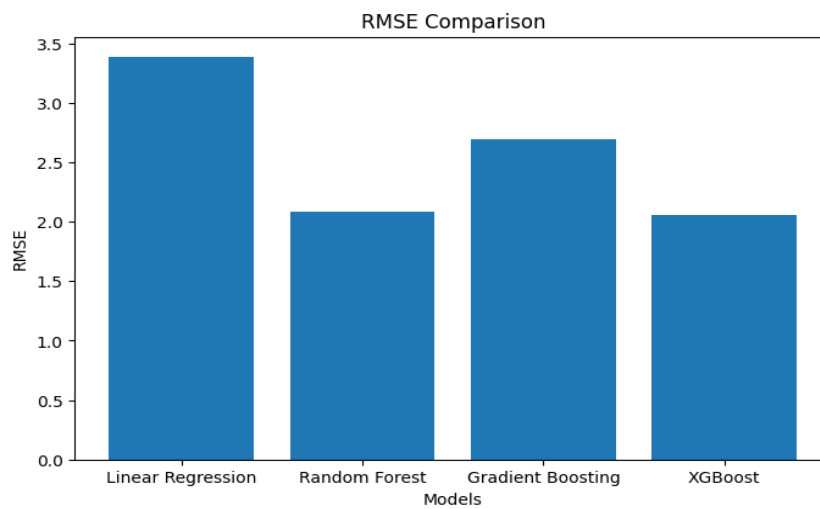


Fig. 3. RMSE Comparison of Models

The RMSE comparison in Fig. 3 shows that XGBoost generated the lowest prediction error among all models trained. Lower RMSE values indicate better prediction accuracy and improved model reliability.

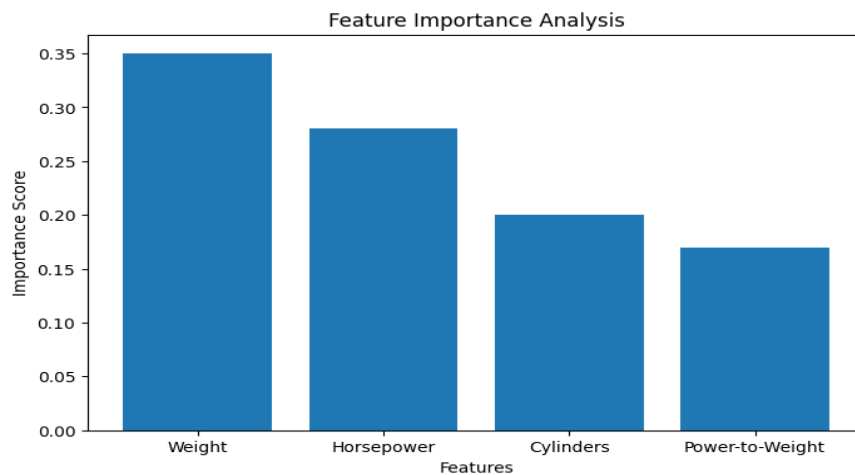


Fig. 4. Feature Importance Analysis

Feature importance analysis in Fig. 4 highlighted that vehicle weight, horsepower, number of cylinders, and power-to-weight ratio were the most influential factors affecting fuel efficiency predictions. The use of Explainable AI techniques improved model transparency by providing insights into how individual features contributed to the predictions.

In addition to predicting fuel efficiency, the proposed framework estimated CO₂ emissions to assess the environmental impact of vehicles. Lower fuel efficiency typically went hand in hand with higher CO₂ emission levels. This underscores the need for efficient vehicle design to promote sustainable transportation.

Fig. 5. Vehicle Configuration Input Interface

The interactive dashboard in Fig. 5 allows users to enter vehicle configuration parameters such as brand, model, horsepower, cylinders, weight, and acceleration for fuel efficiency prediction and analysis.

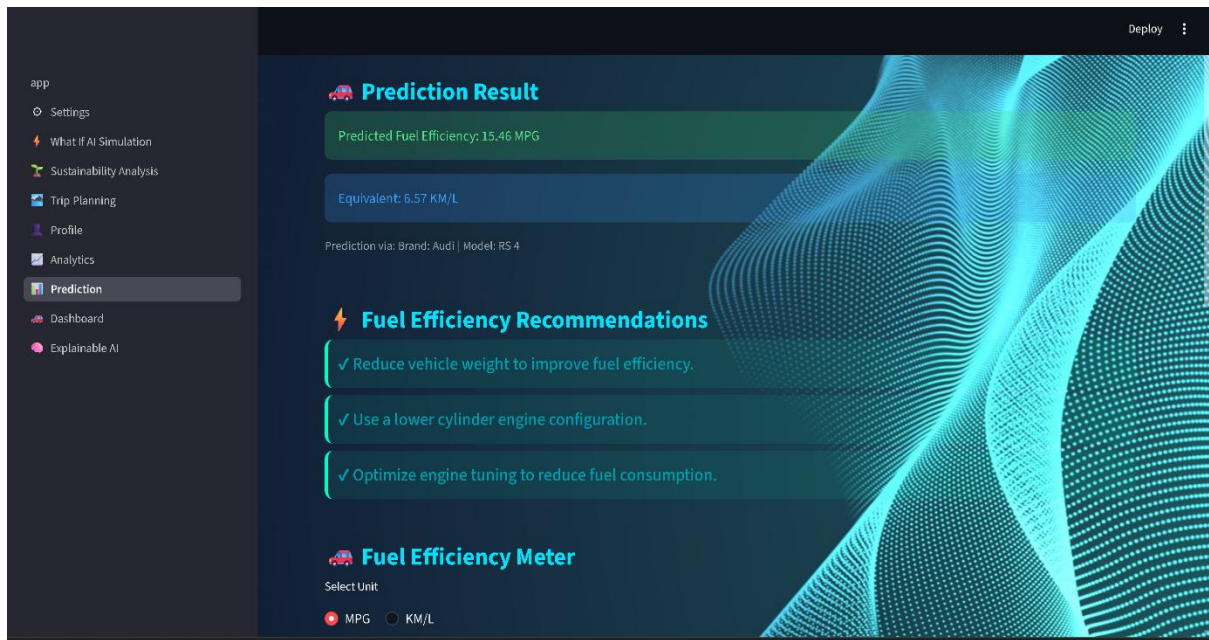


Fig. 6. Interactive Dashboard Prediction Output

The dashboard shown in Fig. 6 provides fuel efficiency predictions along with graphical visualizations and efficiency indicators. Its user-friendly interface supports effective decision-making and practical applications of sustainable vehicle analytics.

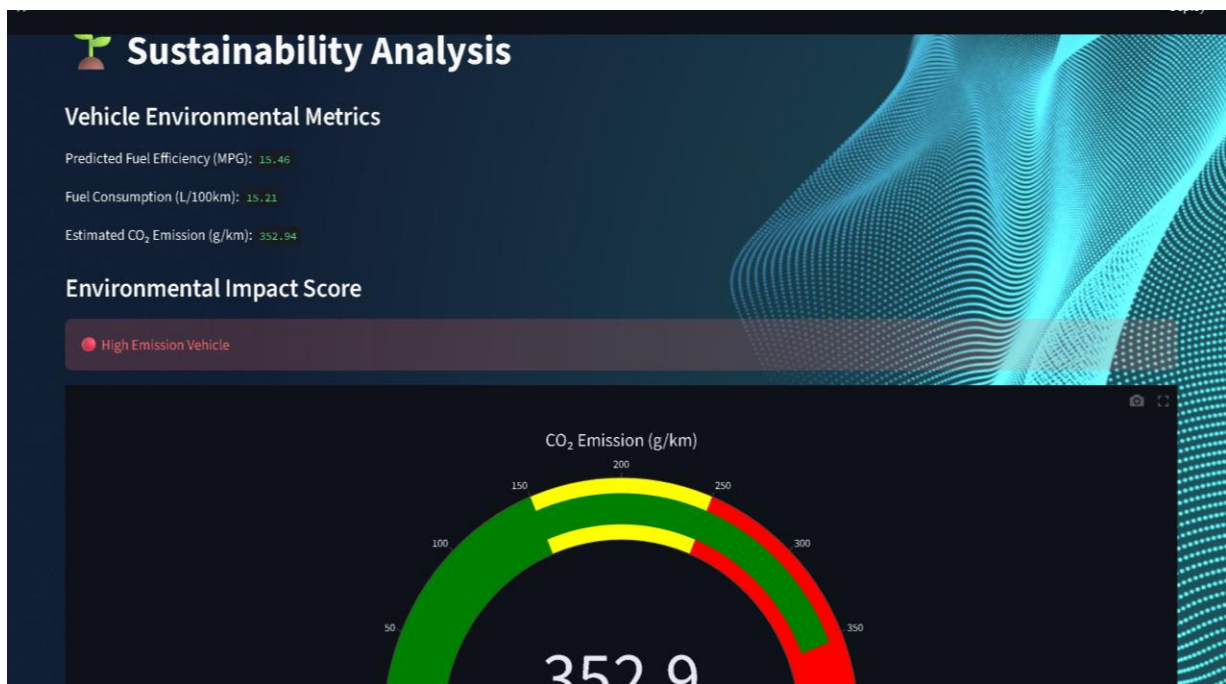


Fig. 7. CO₂ Emission Analysis

The proposed framework estimates CO₂ emissions based on the predicted fuel efficiency of vehicles. This analysis helps assess the environmental effects of vehicle operation and supports sustainability-oriented decision making. The results show that vehicles with lower fuel efficiency tend to produce more CO₂ emissions, while fuel-efficient vehicles lead to a lower environmental impact. This feature

expands the prediction framework by including an evaluation of environmental sustainability alongside performance analysis.

10. Future Scope

The proposed framework can be improved by integrating real-time transportation and navigation services through the Google Maps API. This integration would allow the system to estimate travel distance, expected travel time, and current traffic conditions based on vehicle routes. By combining fuel efficiency predictions with route information, users can get better fuel consumption estimates and make informed travel choices.

In addition, future versions of the system can include IoT-enabled vehicle sensor data for real-time monitoring of engine performance and driving behaviour. Cloud deployment and mobile app integration can increase accessibility and scalability. Also, deep learning methods like Artificial Neural Networks (ANN), Long Short-Term Memory (LSTM), and Transformer-based models may improve prediction accuracy and responsiveness in changing transportation settings.

11. Conclusion

This research proposed an Explainable AI-driven framework for predicting vehicle fuel efficiency using ensemble machine learning techniques and vehicle configuration data. The system included data preprocessing, feature engineering, and several regression-based machine learning algorithms to improve prediction accuracy and model performance.

Experimental results showed that ensemble learning models outperformed traditional regression methods in predicting fuel efficiency. Among all the models tested, XGBoost achieved the highest R^2 score and the lowest RMSE value, demonstrating its ability to manage nonlinear relationships and reduce prediction errors.

The addition of Explainable AI techniques increased model transparency by showing how individual vehicle parameters contributed to prediction outcomes. The framework also estimated CO₂ emissions to assess the environmental impact related to vehicle fuel efficiency, which supports sustainable decision making.

Additionally, an interactive dashboard was created to offer fuel efficiency predictions, graphical visualizations, and simulation-based analysis in an easy-to-use format. Overall, the proposed framework provides an accurate, clear, and practical solution for analyzing vehicle fuel efficiency and promoting sustainable transportation.

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